

Imperial College Deep Learning Project Guidelines (ISIC Lesion Analysis & Report Generation)

This document outlines the complete requirements, technical specifications, precautions, and key deadlines for **Task 1**, **Task 2**, **Task 3**, and the **Bonus Track**.

I. Scoring Breakdown

Assessment Dimension	Weight	Description
Task 1: Segmentation Quality	25%	Dice Score, IoU, 95% Hausdorff Distance (HD95)
Task 2: Attribute Detection	25%	Per-attribute Dice score and Mean Dice across all 5 attributes
Task 3: Report Consistency	20%	JSON Schema compliance, consistency between text, JSON, and masks
Bonus: Retrieval-Augmented Generation	10% (Cap)	OpenCLIP semantic retrieval, RAG report smoothing, and audit trails
Technical Report & Final Presentation	30%	Technical Report (max 12 pages) + Oral Defense

II. Detailed Task Requirements

1. Task 1: Lesion Segmentation

- **Objective:** Accurately segment the skin lesion region (Foreground vs Background).
- **Input:** RGB Dermoscopy Image.
- **Output:** Single-channel binary mask (`.png` format).
- **Evaluation Metrics:**
 - **Dice Coefficient** (Overlap metric)
 - **IoU** (Intersection over Union / Jaccard Index)
 - **95% Hausdorff Distance (HD95)** (Boundary distance deviation / edge accuracy)

2. Task 2: Attribute Detection

- **Objective:** Locate and segment 5 clinically significant dermoscopic patterns.
- **Input:** RGB Dermoscopy Image.
- **Output:** 5 independent binary masks (saved in 5 distinct subdirectories), corresponding to:
 1. `pigment_network`
 2. `negative_network`

3. streaks
4. milia_like_cysts
5. globules

- **Evaluation Metrics:** Per-attribute Dice Score + Mean Dice across all 5 attributes.

3. Task 3: Anchored Findings Report Generation

- **Objective:** Combine predicted outputs from Task 1 and Task 2 to generate a structured JSON report conforming to a strict JSON Schema, alongside a concise human-readable text report.
- **Input:** Task 1 lesion mask + Task 2 attribute prediction logits / masks.
- **Output Files:**
 1. An individual `.json` file for each image.
 2. A consolidated `.csv` file aggregating text reports for all images.

Core Quantitative Mapping Rules (Thresholds & Formulas):

1. Attribute Presence Status:

- Compute mean attribute probability within the Task 1 predicted Lesion ROI:

$$p_{\text{attr}} = \text{mean}(\text{logits}_{\text{attr}}).$$
- Status Determination:
 - present : $p_{\text{attr}} \geq 0.60$
 - absent : $p_{\text{attr}} \leq 0.40$
 - uncertain : $0.40 < p_{\text{attr}} < 0.60$

2. Border Irregularity Category:

- Compute shape irregularity index based on predicted Task 1 Mask:

$$\text{Border Irregularity} = \frac{\text{Perimeter}^2}{4\pi \times \text{Area}}$$

- Edge Case: If $\text{Area} == 0$, set the index to 0.0.
- Classification Rules:
 - irregular : Irregularity Index ≥ 1.60
 - regular : Irregularity Index < 1.60

3. Lesion Size Category:

- Calculate lesion area fraction: Lesion Area Ratio = $\frac{\text{Lesion Pixels}}{\text{Total Pixels}}$
- Classification Rules:
 - small : Ratio < 0.08
 - moderate : $0.08 < \text{Ratio} \leq 0.25$
 - large : Ratio > 0.25

Consistency Checklist (Text & JSON Alignment):

- **Mandatory Medical Terminology:** Generated text reports **must** explicitly mention all 5 attributes (`pigment network` , `negative network` , `streaks` , `milia-like cysts` , `globules`).
- **Strict Status Alignment:** Attribute states in text (present / absent / uncertain) must strictly match the JSON values.
- **Grounded Evidence:** High-confidence attributes must be supported by visual mask evidence to prevent hallucinations.

🌟 Bonus Track: CLIP Semantic Retrieval & RAG Reporting

- **Objective:** Extract embeddings using frozen OpenCLIP models (e.g., `ViT-B/32` or `ViT-L/14`) to retrieve Top-K similar training examples via k NN cosine similarity, smoothing attribute predictions and refining report structure.
- **Output File:**
 - **Bonus Audit CSV:** Containing Query Image ID, Top-K retrieved neighbor IDs, similarity scores, revised text reports, and updated JSONs.
- **Core Principle:** Must enforce **Grounded Outputs**—do NOT blindly copy unverified pathological features from neighboring images into the current report!

⚠️ III. Critical Technical Guidelines & Pitfalls

1. **Interpolation Methods during Data Augmentation (Crucial):**
 - **RGB Images:** Use **Bilinear Interpolation** when resizing or cropping.
 - **Binary Masks:** **NEVER** use bilinear interpolation! **Must** use **Nearest-Neighbor Interpolation** to avoid non-binary border artifacts.
2. **Data Alignment & Quality Control (QC):**
 - Ensure geometric transformations (Flip, Rotate, Crop) are applied synchronously across images and all corresponding masks.
 - Masks must strictly maintain binary values (0/1 or 0/255).
3. **Task 2 Class Imbalance:**
 - Sparse features like `streaks` appear infrequently.
 - Use composite loss functions (e.g., `BCE Loss + Dice Loss`), and evaluate attributes individually during validation.
4. **Experimental Reproducibility:**
 - Set global random seeds.
 - Sort Image IDs before shuffling for the 80% Train / 20% Val split, and save partition details into `split.csv` .
5. **Technical Report Constraints:**
 - Required sections: Literature Review, Methods, Results and Discussion, Future Work.
 - Length limit: **Maximum 12 pages** (excluding References).

📦 IV. Deliverables Checklist

Each group needs to submit one complete Deliverables package:

1. **Task 1 Lesion Masks:** PNG format (one file per image).
2. **Task 2 Attribute Masks:** PNG format (stored in 5 separate directories named by attribute).
3. **Task 3 Findings Report:**
 - Individual `.json` file for each image.
 - Combined `.csv` file with text reports for all images.
4. **Bonus Audit File (Optional):** CSV file detailing Top-K neighbor IDs, modified reports, and updated JSONs.
5. **Technical Report & Code:** Technical Report PDF + well-commented, reproducible codebase.

V. Important Schedule & Deadlines

Date	Time	Event / Milestone
July 21	-	Technical Tutorial Released
July 27	-	Submit Literature Review (Lesion Segmentation)
July 30	13:00 (1 PM)	Test Set Released
July 30	19:00 (7 PM)	Final Submission Deadline: Code, Predictions & Technical Report
July 31	-	Final Presentations & Award Ceremony (Best Overall & Best Segmentation Model)